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Enhanced Recovery After Surgery 2.0: Optimizing Pain Management in Nuss Procedure: Cryoablation and Nerve Block Strategies for Reduced Opioid Use

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ABSTRACT

Introduction: Our study assesses the association between cryoablation, with and without nerve block supplementation, post-Nuss procedure pain, and opioid use in pectus excavatum (PE) patients.

Methods: We conducted a retrospective cohort study at a single center for PE patients who underwent the Nuss procedure from 2017 to 2022. Outcomes included postoperative opioid use (measured in oral morphine milligram equivalent per kilogram [OME/kg]), average pain score (scale 0-10), and length of stay (LOS).

Results: One hundred sixty-four patients (146 males and 18 females) were included, with 79 (48.2%) receiving neither cryoablation nor nerve block, 60 (36.6%) receiving intraoperative cryoablation alone, and 25 (15.2%) receiving both cryoablation and nerve block. The median age was 16 y. Nerve block recipients consumed fewer opioids during hospitalization than cryoablation alone and nonintervention groups (1.5 *versus* 2.3 *versus* 5.8 OME/kg, respectively, $P < 0.0001$). Average pain scores over the total LOS were lower in nerve block recipients (3.5 *versus* 3.8 *versus* 4.2, $P = 0.03$), particularly on postoperative day 0 ($P = 0.002$). Nerve block recipients had a shorter LOS than cryoablation alone and nonintervention groups (43.4 *versus* 54.7 *versus* 66.2 h, $P < 0.0001$). On multivariate analysis, cryoablation alone resulted in significantly less opioid use compared to no intervention (3.32 OME/kg reduction, 95% confidence interval −4.16 to −2.47, $P < 0.0001$). Addition of nerve block further reduced opioid use by 1.10 OME/kg (95% confidence interval −2.07 to −0.14, $P = 0.04$).

Conclusions: Cryoablation with nerve block supplementation is associated with reduced pain, opioid use, and LOS post-Nuss for PE repair compared to cases without cryoablation or with cryoablation only. Cryoablation with regional nerve blocks should be considered for Nuss repair under the enhanced recovery after surgery pathway.

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Introduction

Pectus excavatum (PE) is a congenital birth defect that becomes increasingly apparent during adolescent growth. It is the most common chest wall deformity, estimated to affect 1 in 400 live births with a higher prevalence in males to females (5:1).[1-4] The Nuss procedure provides a minimally invasive thoracoscopic approach to the repair of PE and is currently the standard of care in the United States.[5,6] However, postoperative pain management following the Nuss procedure remains challenging. Over the years, multiple pain modalities were employed, such as thoracic epidurals, intravenous patient-controlled analgesia (PCA) with opioids, long- and short-term regional nerve blocks, oral opioids, and thoracoscopic intercostal nerve cryoablation (INC).[6-10] While opioids are effective for pain management, they are associated with a range of side effects including pruritus, nausea, vomiting, constipation, urinary retention, sedation, and respiratory depression.[11] Postoperative use of opioids can increase the possibility of dependence and misuse, especially in the vulnerable adolescent population.[12,13] Opioid-sparing protocols are needed following the Nuss procedure while still maximizing pain control and avoiding opioid side effects. Application of perioperative standardization of care to improve surgical outcomes after the Nuss procedure, such as the enhanced recovery after surgery (ERAS), has been shown to improve functional mobility, minimize infection, and encourage opioid-sparing analgesia.[14-16]

Cryoanalgesia is a valuable modality due to its analgesic effect lasting throughout the postoperative recovery period. Intercostal nerve cryoablation has been increasingly used for postoperative pain control following the Nuss procedure and has been shown in multiple studies to decrease length of stay (LOS) and opioid usage.[6,9,17,18] Using the cryoprobe releases high-pressure carbon dioxide or nitrous oxide, rapidly cooled to −50°C to −70°C, thereby causing degeneration of the nerve axons and temporarily preventing pain transmission for several weeks to months.[17-19] Sensation returns as the nerve axons regenerate within the intact nerve sheath.[11]

Anesthesia-performed peripheral nerve blocks such as erector spinae plane block (ESPB) and thoracic paravertebral blocks (TPVBs) have gained popularity for postoperative pain control following the Nuss procedure. ESPB is an interfascial plane block where local anesthetics are injected superficially into the transverse process and beneath the iliocostalis, longissimus, and spinalis muscles, thereby avoiding damage to peripheral nerves and blood vessels.[8,17-19] ESPBs diffuse into the paravertebral space, blocking the dorsal and ventral branches of the thoracic spinal nerves covering the dermatome.[17] In TPVBs, local anesthetic is injected lateral to the thoracic vertebra close to spinal nerves emerging from the intervertebral foramina, giving nerve blockade to the thoracic dermatomes above and below the site of injection.[20] TPVBs have practical analgesic effects similar to thoracic epidurals but require a high level of proficiency due to the risk of pneumothorax and neurovascular damage.[18]

Our institution sought to improve the current ERAS protocol for patients undergoing the Nuss procedure by reducing postoperative pain management through the use of INC. The previous ERAS Nuss protocol at this institution was established in 2017 and found a reduction in hospital LOS and a decrease in urinary catheter usage without an increase in postoperative emergency department visits and hospital readmissions.[14] Our practice is distinct from other institutions' because it includes preoperative components such as 3 d of loading with gabapentin, bowel preparation, extensive preoperative stretching, and particular emphasis on preoperative teaching and counseling to set patient and parental expectations by a dedicated nurse practitioner team who also follows the patients after discharge.[14] Patients are managed jointly postoperatively by both surgery and the pediatric acute pain service. Our current study sought to assess the efficacy of the ERAS pathway in our institution from 2017 to 2022 after different pain analgesia strategies were implemented. In 2019, we added cryoablation to our pathway, and then in 2022, nerve blocks were added. Our study aims to evaluate the association of cryoablation, with and without nerve block supplementation, on post-Nuss procedure pain, opioid use, and LOS in patients with pectus excavatum. We hypothesize that the addition of peripheral nerve blocks along with cryoablation will be associated with a decrease in pain levels, opioid use, and LOS.

Materials and Methods

We conducted a retrospective review of all patients under 21 y of age who underwent Nuss procedure for surgical correction of PE from 2017 to 2022 at a single academic institution. Patients who underwent the Ravitch procedure for PE repair were not included in this study. We also excluded one patient who underwent sternotomy for cardiac repair concurrently with the Nuss procedure. Additionally, a patient with myasthenia gravis was excluded due to the use of epidural anesthesia instead of opioid anesthesia. All patients received a single pectus bar. Institutional review board (IRB00158253) approval was obtained prior to conducting this study. Individual patient consent was not sought, as the study operated under an institutional review board-approved waiver of consent in compliance with the Health Insurance Portability and Accountability act.

Patients were divided into three groups based on the standardized ERAS pathway that was in place on the date of operation (January 2017-November 2019: no cryoablation; December 2019-2021: cryoablation alone; and January-December 2022: cryoablation + peripheral nerve block). Age, Haller index (HI), gender, postoperative pain scores, inpatient opioids administered, postoperative LOS, hospital readmissions, and emergency department visits were recorded. The primary outcome of interest was inpatient postoperative oral morphine milligram equivalent per kilogram (OME/kg) usage. OMEs were calculated using conversion factors derived from international sources.[21] Secondary outcomes included patients' self-reported postoperative pain scores on a scale of 0 (no pain) to 10 (excruciating pain), as well as postoperative LOS, hospital readmission, and emergency department visits related to the surgical procedure.

ERAS pathway without cryoablation

Patients undergoing the Nuss procedure from January 2017 to November 2019 were included in the ERAS Nuss protocol without cryoablation. Once identified as a surgical candidate, the patients underwent extensive and standardized outpatient education by a pediatric nurse practitioner regarding preoperative preparation, expectations for surgery, and postoperative benchmarks.[14] Outpatient preoperative medications of gabapentin dosed nightly and polyethylene glycol dosed twice a day were initiated 3 d before surgery and were successfully completed by all patients in our study. Gabapentin was titrated three times a day postoperatively, then weaned over 3 wk. The use of gabapentin effectively reduces acute postoperative pain, is opioid sparing, has a lower incidence of chronic postsurgical pain, and is associated with lower levels of postoperative anxiety and emergence agitation in pediatric patients.[22,23]

Before discharge, surgical nurse practitioners counseled family members using a postsurgical handbook detailing activity restrictions and conditions requiring discussion with the surgical team. A 3-wk medication regimen consisting of standing acetaminophen and ibuprofen, opioid wean (oxycodone or hydromorphone), gabapentin weans, prophylactic antacid during nonsteroidal anti-inflammatory drug usage, and a bowel regimen to avoid constipation was prescribed. Diazepam was prescribed as needed for treatment of muscle spasms. In addition to the handbook, patients were given 12-wk physical therapy guidance to advance their activity, developed from a collaboration between pediatric surgery and physical therapy.

ERAS pathway with cryoablation

Patients undergoing the Nuss procedure from December 2019 to December 2021 were included in the ERAS pathway with cryoablation. The standard ERAS Nuss pathway was used, and the pain regimen for discharge was adjusted to account for cryoablation. Once ready for discharge, patients were instructed to take standing acetaminophen and ibuprofen for 1 wk, gabapentin twice a day for 7 d, then nightly for 90 d at the recommendation of the cryoablation manufacturer in order to prophylactically mitigate neuropathic pain[24] and provided a 3-d supply of oxycodone or hydromorphone to take as needed for breakthrough pain at home. Diazepam was prescribed as needed if patients complained of muscle spasms or chest tightness.

ERAS pathway with cryoablation with peripheral nerve blocks

Patients undergoing the Nuss procedure in 2022 were included in the ERAS pathway with cryoablation with peripheral nerve block. Anesthesia is placed on a single-shot ESPB or TPVB prior to surgical incision. Discharge regimen was the same as the ERAS pathway with cryoablation.

Intraoperative pain control

All three groups received methadone 0.1-0.15 mg/kg (up to 10 mg) IV and ketorolac 0.5 mg/kg (up to 30 mg) IV. The no cryoablation group received intraoperative three-level bilateral intercostal nerve blocks and a field block around the incisions using up to 1 mL/kg of 0.25% bupivacaine. A thoracoscope was placed in the thoracic cavity to monitor for ice crystal formation using the CryoICE cryoablation probe. The probe was cooled to -60°C for 2 min.[9] The probe was warmed to ambient temperature and allowed to release from the nerve without traction. This procedure was completed consistently through the third to the seventh intercostal spaces bilaterally in all patients.

Under aseptic conditions, ESPBs were placed after induction of general anesthesia and prior to incision. Linear, high-frequency ultrasound guidance was used in real-time to place the bilateral single-shot nerve blocks at the T4-T6 level with up to 20 mL of 0.35% ropivacaine and 0.5 mcg/kg of clonidine, using a paramedian approach. In a similar manner, bilateral ultrasound-guided TPVBs were placed after induction of general anesthesia and prior to incision. The patient's back was sterilely prepped and draped after positioning in lateral decubitus position and the T4-T6 level along with surrounding structures were identified. Under real-time ultrasound guidance, the paravertebral space was identified, and the insertion of the needle tip was confirmed until just past the costotransverse ligament. Approximately 20-25 mL of the mixture of 0.35% ropivacaine and 0.5 mcg/kg of clonidine were injected in 3-5 mL increments after negative aspiration and with appropriate inferior displacement of the pleura. The procedure was then repeated on the contralateral side at the same level.[19]

Postoperative pain management

Patients were enrolled in patient-report-driven multimodal pain management protocols. Per institutional ERAS, all patients started with PCA of either hydromorphone or morphine, gabapentin, scheduled acetaminophen and ketorolac, intermittent benzodiazepine (diazepam), and a low-dose ketamine infusion (0.1 mg/kg/h). While in the post-anesthesia care unit, the opioid PCA was on basal rate with dosing at the discretion of the anesthesiologist in the no cryoablation group, and once cryotherapy was employed, the opioid PCA was on bolus only with dosing per anesthesia. All patients in each group were transitioned to scheduled ibuprofen, acetaminophen, an oral opioid (oxycodone or hydromorphone), and a benzodiazepine for muscle spasms as needed. IV analgesia was usually weaned 1-2 d postoperatively, depending on the patient's pain tolerance, pain control scores, PCA usage, and ability to tolerate enteral medications.

Discharge readiness

Patients were deemed ready for discharge when they met all the following criteria: pain adequately controlled by oral analgesia, no active emesis, ambulating without assistance, and family expressed readiness and capability to manage recovery at home. The surgical nurse practitioner discharged the patient and family with the Nuss postsurgery handbook, detailing medication side effects, activity restrictions, physical therapy guidelines, and advancement activity, and when to contact pediatric surgery or return to the pediatric emergency department.

Statistical analysis

We conducted a comparative analysis to assess the differences in characteristics and outcomes among patient groups. Categorical variables, presented as frequency and percentage, were compared with Pearson's chi-squared or Fisher's exact tests. For continuous variables, normality was assessed using the Shapiro–Wilk test. Non-normally distributed variables and ordinal variables, presented as median (interquartile range), were compared using the Kruskal–Wallis H test, while normally distributed variables, presented as mean ± standard deviation, were analyzed using the analysis of variance. To explore the association between treatment modality and total OME/kg given during the hospital stay, multivariable linear regression was performed. The final model included Haller index, sex, and age as potential confounding factors. A bivariate linear regression was utilized to illustrate the regression coefficient between the Haller index and total OME/kg. Statistical significance was defined as a P value < 0.05. All statistical analyses were conducted using R version 4.2.2 (R Core Team, 2022).

Results

Between January 2017 and December 2022, a total of 166 patients underwent the Nuss procedure for surgical correction of pectus excavatum (PE). Two patients were excluded due to epidural pain management and concurrent cardiac surgery, leading to a cohort of 164 patients (Appendix 1). One hundred forty-six (88.0%) patients were male, while 18 (12.0%) were female with a median age of 16 y (interquartile range 15-17) at the time of surgery. The patients were divided into three groups based on the year they had surgery, which reflected which intervention they received: 79 (48.2%) received the ERAS Nuss pathway without cryoablation, 60 (36.6%) received the ERAS with cryoablation alone, and 25 (15.2%) received the ERAS with cryoablation and bilateral pain block (ESPB or TPVB). Baseline characteristics including age, sex, weight, and Haller index were similar between the three groups (Table 1).

Primary outcome

Patients who received cryoablation alone or cryoablation with nerve block consumed significantly fewer opioids during their hospitalization compared to those who did not receive cryoablation (median 2.0 *versus* 1.0 *versus* 5.5 OME/kg, respectively; $P < 0.0001$) (Table 2). Notably, they exhibited significantly lower daily opioid consumption on postoperative days (PODs) 0, 1, 2, and 3 ($P < 0.0001$) (Fig. 1, Appendix 2). In the cryoablation group, one patient with Marfan syndrome experienced bilateral pleural effusion, necessitating bilateral chest tubes insertion and an extended hospitalization period of 12 d. Thus, the spike in opioid usage observed on POD 5 in Figure 1 can be directly attributed to this specific patient, as they were the only individual in the group with a postoperative LOS exceeding 4 d.

On multivariable analysis, cryoablation significantly reduced opioid use by 3.32 OME/kg (95% confidence interval [CI] −4.16 to −2.47) compared to no cryoablation. Additionally, the combination of cryoablation with nerve block reduced opioid use by 4.40 OME/kg (95% CI −5.52 to −3.27) throughout the entire hospital stay. Moreover, when comparing cryoablation alone to cryoablation with nerve block, the latter reduced opioid use by 1.1 OME/kg (95% CI −2.07 to −0.14) (Table 3).

Secondary outcomes

The median daily patient reported pain scores on the day of surgery (POD 0) differed significantly among the no cryoablation group, cryoablation alone group, and cryoablation with nerve block group (4.6 *versus* 4.0 *versus* 2.8, respectively, $P = 0.003$) (Fig. 2, Appendix 3). Furthermore, patients that did not receive cryoablation reported more pain during the entire hospital stay compared to those treated with cryoablation alone and cryoablation with nerve block (median 4.2 *versus* 3.7 *versus* 3.3 d, respectively, $P = 0.04$) (Table 2). At initial follow-up at 2-4 wk postoperatively, all patients demonstrated paresthesias in areas of thoracic distribution of cryotherapy. By 3-mo follow-up, all patients demonstrated resolution of paresthesia by self-report.

Nerve block recipients had a shorter postoperative LOS than cryoablation alone and nonintervention groups (mean 41 *versus* 53 *versus* 65 h, $P < 0.0001$). Nevertheless, the duration of surgery was longer in both cryoablation and cryoablation plus nerve block groups (both 130 min) compared to the no cryoablation group (72 min) ($P < 0.0001$). No patients who received cryoablation with nerve block were readmitted to the hospital for pain, while one (1.7%) patient who received cryoablation alone and two who did not (2.5%) were readmitted ($P = 1$).

Table 1 – Demographic characteristics of patients undergoing Nuss procedure by pain management group.

Demographics	Overall, $n = 164$	No cryoablation, $n = 79$	Cryoablation, $n = 60$	Cryoablation + nerve block, $n = 25$	P value
Age (y), median (IQR)	16 (15-17)	16 (15-17)	16 (15-17)	16 (15-16)	0.57
Sex, *n* (%)					0.51
Female	17 (10.3)	10 (12.7)	5 (8.3)	3 (12.0)	
Male	146 (89.7)	69 (87.3)	55 (91.7)	22 (88.0)	
Weight (kg), mean ± SD	58.2 ± 8.9	59.5 ± 9.9	56.5 ± 7.7	57.8 ± 8.4	0.15
Haller index, mean ± SD	4.7 ± 1.4	4.6 ± 1.4	4.9 ± 1.6	4.8 ± 1.0	0.36

IQR = interquartile range; SD = standard deviation.

Table 2 – Primary and secondary outcomes of patients undergoing Nuss procedure by pain management group.

Outcomes	No cryoablation, $n = 79$	Cryoablation, $n = 60$	Cryoablation + nerve block, $n = 25$	P value
Total OME/kg, median (IQR)	5.5 (4.0-7.2)	2.0 (0.8-3.3)	1.0 (0.4-1.6)	**<0.0001**
Total pain score, median (IQR)	4.2 (3.4-5.0)	3.7 (2.9-4.8)	3.3 (2.2-4.6)	**0.04**
Postoperative LOS (h), mean ± SD	65 ± 20	53 ± 31	41 ± 14	**<0.0001**
Length of surgery (min), mean ± SD	72 ± 21	130 ± 25	130 ± 25	**<0.0001**
Readmission for pain, *n* (%)	2 (2.5)	1 (1.7)	0 (0)	1
Readmission*, *n* (%)	3 (3.8)	3 (5.0)	1 (4.9)	1

P values in bold represent statistical significance, $P < 0.05$.
LOS = length of stay; OME = oral morphine equivalent; IQR = interquartile range; SD, standard deviation.
*Reasons for readmissions include fever, ileus, pericardial effusion, and pneumothorax.

On multivariate analysis, we found that a higher Haller index (HI) was significantly associated with increased opioid use (beta coefficient: 0.31, 95% confidence interval: 0.04 to 0.59, $P = 0.03$). To gain further insight into this association, we conducted a subanalysis, as depicted in Figure 3. When all patients were taken into consideration, there was no association between Haller index and total OME/kg (regression coefficient [R] = 0.095, $P = 0.23$) (Fig. 3A). Moreover, when looking within the no cryoablation group, a linear relationship was observed between a higher HI and increased total OME/kg ($r = 0.25$, $P = 0.002$) (Fig. 3B). However, this association disappeared among patients treated with cryoablation, with or without nerve block ($r = -0.0006$, $P = 1$) (Fig. 3C).

Discussion

Postoperative analgesia after the Nuss procedure is challenging but essential for acceptable postsurgical outcomes. After initial implementation of our ERAS pathway in 2017, we noted decreased LOS, decreased Foley usage, and readmissions. Our institution wanted to further improve pain management after this initial success by incorporating cryoablation in 2019. We found improved pain management and decreased usage of opioids, as well as a reduction in postoperative LOS. We subsequently added peripheral nerve blocks to our standardized pathway in 2021. After the addition of cryoablation, gabapentin dosing in the postoperative period was adjusted based on recommendations from our anesthesiologists and manufacturer of the cryoprobes. Therefore, the group without cryoablation is our historical group that we are using for comparison to demonstrate the effectiveness of our new ERAS modifications.

Our data suggests that using the ERAS pathway with cryoablation is associated with decreased LOS, inpatient OME/kg, and inpatient pain scores compared to historical controls. Cryoablation leaves the myelin sheath and endoneurium intact and causes transient paresthesia that diminishes the patient's pain response.[7] The total effect of cryoablation can take up to 36 h to take full effect. The cryoprobe manufacturer

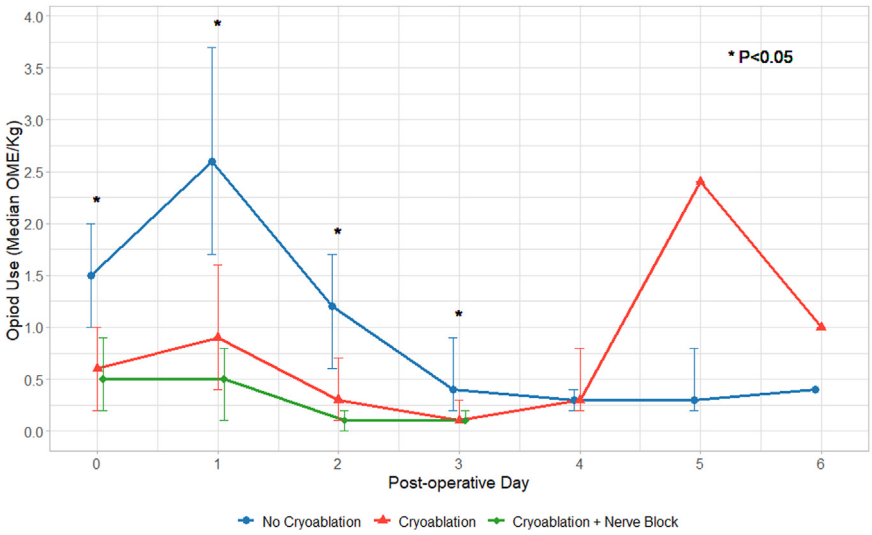

Fig. 1 – Median OME/kg per postoperative day in children across three treatment groups: no cryoablation (blue circles), cryoablation alone (yellow triangles), and cryoablation with nerve block (green squares). Data are presented as median OME/kg per postoperative day. Error bars represent the interquartile range. The Kruskal–Wallis H test was used to compare the groups. * Indicates a *P* value < 0.05, which was considered statistically significant.

Table 3 – Multivariate linear regression analysis for OME/kg, controlling for age, sex, and Haller index.

Opioids (OME/kg)	β-Coefficient	95% confidence interval	*P* value
Pain control			
No cryoablation	Reference		
Cryoablation	−3.32	−4.16 to −2.47	**< 0.0001**
Cryoablation + nerve block	−4.40	−5.52 to −3.27	**< 0.0001**
Pain control*			
Cryoablation	Reference		
Cryoablation + nerve block	−1.10	−2.07 to −0.14	**0.03**
Haller index	0.31	0.04 to 0.59	**0.03**
Sex			
Female	Reference		
Male	−1.38	−2.81 to −0.35	**0.01**
Age (y)	−0.06	−0.24 to 0.13	0.69

OME = oral morphine equivalent.
*This shows the result of the multivariate linear regression with cryoablation alone, which was used as a reference in order to compare it to cryoablation + nerve block. *P* values in bold represent statistical significance, $P < 0.05$.

describes nerve function returning in 1-3 mo, which is consistent with other reported data showing 3 mo for younger patients (<21 y of age) and >3 mo for older patients >21 y of age.[16,24-26] All patients in our study endorsed paresthesia resolution within 3 mo following surgery, consistent with previously reported data.[24] Timing of resolution of postoperative paresthesia was not tightly followed when cryotherapy was employed. All patients had paresthesia at their first clinic follow-up appointment 2-4 wk postoperatively and were not seen again unless there are clinical concerns until their 12-wk follow-up appointment, in which all patients reported that paresthesia had resolved.

Regional anesthesia using ESPB or TPVB with cryoablation was further associated with a decreased LOS (from 54.7 h to 43.4), inpatient OME/kg (from 2.0 to 1.0), and inpatient pain scores (from 3.7 to 3.3). The primary outcome measure was the difference in inpatient total OME/kg, which is objectively measured and carries great importance from a clinical perspective. Comparison of total OME/kg across all three groups showed that cryoablation with nerve block was successful at decreasing the total opioid administered to patients during the hospital stay. Limiting opioid consumption reduces postsurgical complications associated with opioid usage, such as nausea, vomiting, constipation, and respiratory depression.[8] Additionally, reducing opioid usage is beneficial for lowering addiction risk among at-risk adolescents.[8,12,13]

Secondary outcomes examined were averaged inpatient pain scores, LOS and readmission, and emergency department

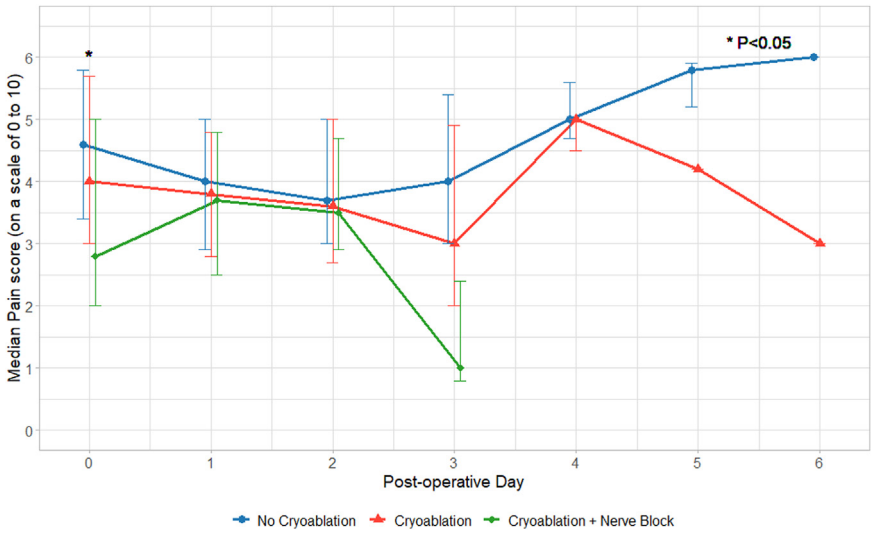

Fig. 2 – Median pain scores (scale 0-10) per postoperative day in children across three treatment groups: no cryoablation (blue circles), cryoablation alone (yellow triangles), and cryoablation with nerve block (green squares). Data are presented as median pain score on a scale of 0-10 per postoperative day. Error bars represent the interquartile range. The Kruskal–Wallis H test was used to compare the groups. * Indicates *P* value < 0.05, which was considered statistically significant.

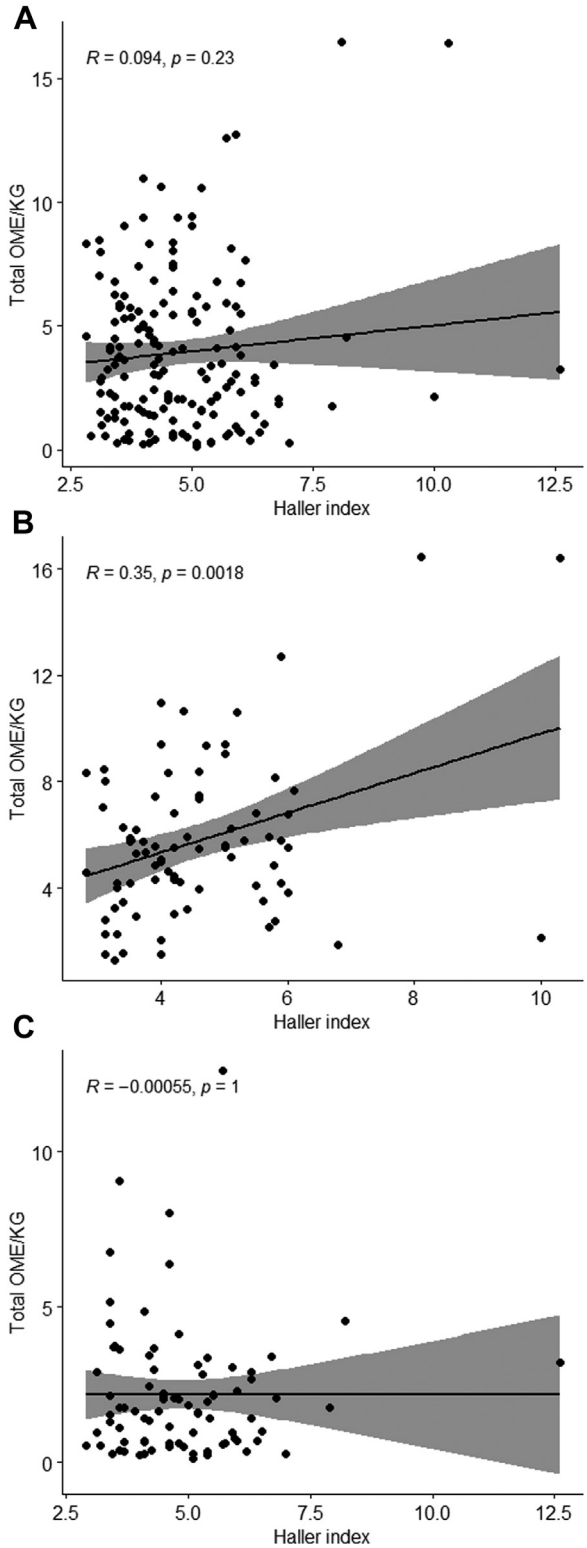

Fig. 3 – Relationship between Haller index and total OME/kg. The scatter plot shows the relationship between Haller index and total OME/kg. Each data point represents an individual patient. A regression line with a confidence interval illustrates the trend between the variables. The regression coefficient (R) indicates the change in total OME/kg associated with a one-unit increase in the Haller index, while the displayed *P* value signifies the statistical significance of the coefficient. The figure includes three patient plots: (A) includes the entire study cohort; (B) consists of patients without cryoablation; and (C) comprises patients who received cryoablation ± nerve block. A *P* value of < 0.05 was considered statistically significant.

visits. Assessing the efficacy of analgesia techniques is complex, and numeric pain rating score has limitations because pain is a subjective experience and pain perception varies among patients.[8,27] Numeric pain rating score can be more helpful in comparing pain experience in a specific patient and perhaps less valuable between patients.[27] Our study revealed that patients who received cryoablation with nerve block experienced significantly lower pain scores on POD 0 and throughout their entire hospital stay (2.8 on POD 0, 3.3 total) compared to those treated with cryoablation alone (4.0 on POD 0, 3.7 total). Notably, patients in the no cryoablation group reported the highest pain scores (4.6 on POD 0, 4.2 total). These findings highlight the beneficial impact of adding a nerve block to cryoablation, effectively reducing short-term postprocedural pain.

Cryoablation offers a significant advantage by reducing postoperative LOS by nearly a day (3 d in cryoablation *versus* 4 d in no cryoablation), which is a finding supported by other studies.[6,7,16,25] Other groups have reported decreased LOS to between 2 and 4 d.[9,25,28-30] Five studies specifically looked at comparing outcomes using thoracoscopic cryotherapy to thoracic epidural and demonstrated reductions in LOS from 5 to 6 d with epidurals to 2 to 3 d with cryotherapy.[9,25,28-30] A study recently published on the use of INC and bupivacaine blocks noted that out of a series of 40 patients (92.5%) were discharged home the day after surgery, and 7.5% of patients were discharged on POD 2 for an overall LOS of 1.2 d.[16] It is important to note that cryoablation requires a longer operative time, as expected, since each side requires at least 30 min for ablating 5 intercostal nerves (T3-T7). Due to nerve blocks occurring prior to skin incision, this added procedural time was not factored into our operative time. Prolonged surgery duration is typically associated with increased surgical costs. Therefore, conducting a cost analysis would provide valuable insights into how cryoablation impacts the overall cost of hospitalization in Nuss procedures. The added procedural time for a nerve block would also need to be considered in future cost analyses.

With a standardized ERAS pathway among all three groups, there was no difference in hospital readmissions or ED visits. No cryoablation with nerve block patients were readmitted, while one patient who received cryoablation alone and two who did not receive cryoablation were readmitted, further supporting the efficacy of these approaches. With the added benefit of nerve block supplementation, patients met discharge criteria sooner (43.4 ± 13.9 h to discharge) than patients who did not receive cryoablation (66.2 ± 19.4 h) or patients with cryoablation alone (54.7 ± 31 h). Our institution concluded that using cryoablation with nerve blocks for the standard ERAS pathway for the Nuss procedure is superior to the ERAS cryoablation without nerve blocks. From our institution's perspective, all Nuss procedures currently include cryoablation with ESPB or TPVB.

To date, there has been limited research on the impact of Haller index (HI) on pain. In our study, we observed that in the no cryoablation group, a higher HI was correlated with increased OME/kg ($r = 0.35$). Patients treated with cryoablation, with or without nerve block, showed no association between HI and OME/kg ($r = -0.00055$). These findings reinforce our conclusion that cryoablation effectively reduces pain in patients with a higher HI, particularly those who are more prone to experience increased pain.

The study is limited by its retrospective design. Our data are drawn from a single large academic institution from the practice of seven general pediatric surgeons. Another critique of this study is a lack of standardization between groups that underwent ESPB ($n = 19$) or TPVB ($n = 6$). Few patients received TPVB during this study, and TPVB *versus* ESPB was used at the anesthesiologist's discretion with TPVB initiated in our institution toward the end of 2022. Furthermore, our study is not sufficiently powerful to show the true difference between the cryoablation alone group and the cryoablation + nerve block group due to the small sample size. Nevertheless, the positive trend is worth noting, and future studies should evaluate the impact of cryoablation with ESPB or TPVB supplementation. A future addition to this study design would be inclusion of patient and parent satisfaction with pain management.

Conclusions

Cryoablation is associated with reduced pain, opioid use, and LOS post-Nuss for pectus excavatum repair within an ERAS pathway. Nerve block supplementation with cryoablation showed shorter LOS and decreased inpatient opioid usage and therefore should be considered for postoperative pain management. Future studies will look at a standardized approach in deciding which nerve block supplementation (ESPB or TPVB) along with cryoablation provides optimal postoperative pain management for patients undergoing the Nuss procedure for pectus excavatum.

Supplementary Materials

Supplementary data related to this article can be found at https://doi.org/10.1016/j.jss.2024.07.003.

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Kristin Wharton: Writing – review & editing, Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Charbel Chidiac:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Carla Lopez:** Writing – original draft, Investigation, Formal analysis. **Joann Hunsberger:** Writing – original draft, Data curation, Conceptualization. **Daniel Rhee:** Writing – review & editing, Methodology, Investigation. **Clint Cappiello:** Writing – review & editing, Methodology, Investigation. **Alejandro V. Garcia:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Investigation, Conceptualization.

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